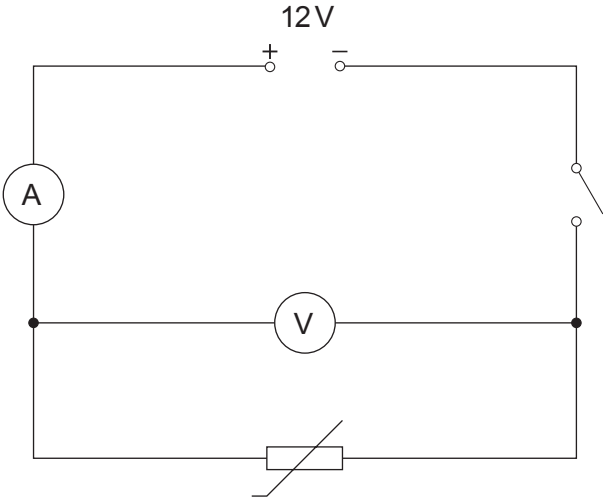


4. A thermistor is a type of resistor whose resistance depends on temperature. Thermistors are often used as temperature sensors in applications where we want to monitor or control temperature. A group of students immerse a thermistor and a thermometer in water in order to determine the resistance at different temperatures. The circuit they use is shown below.



Their results are given below.

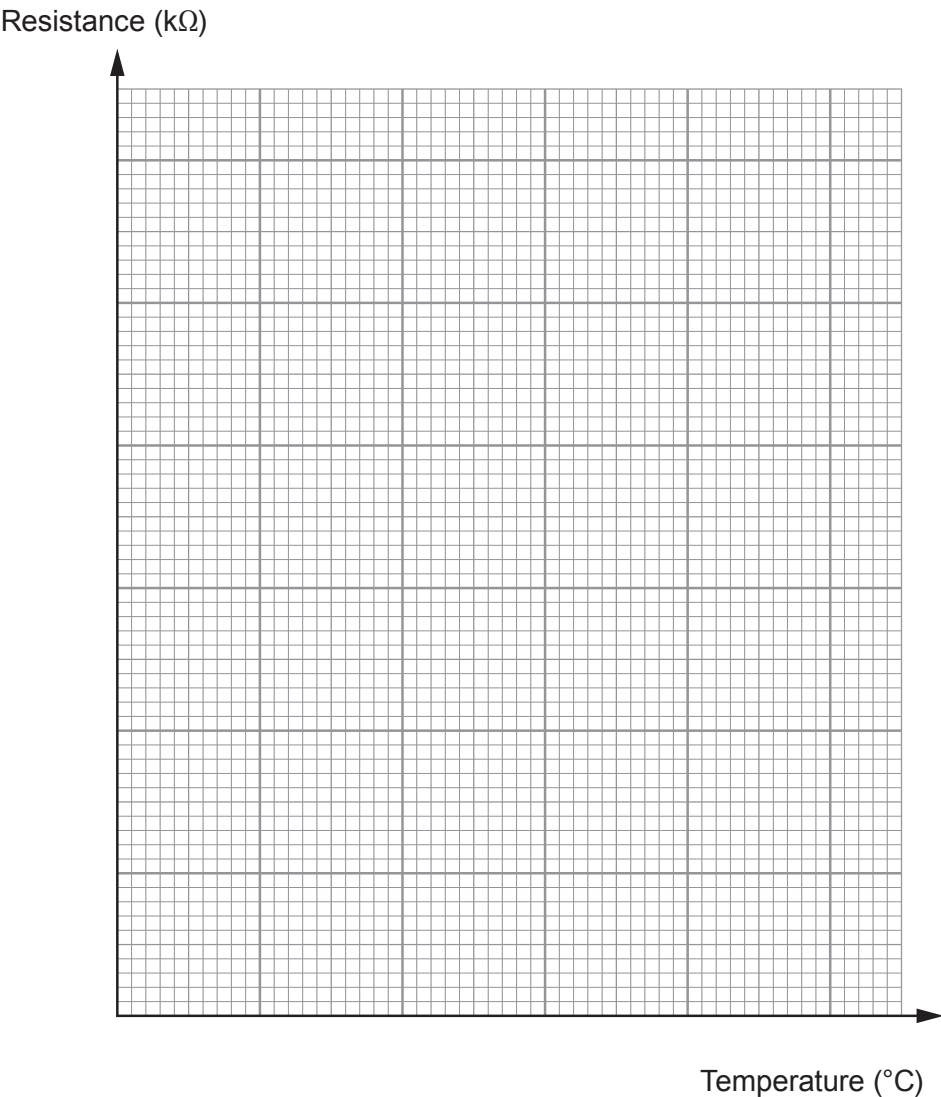
Temperature (°C)	Current (mA)	Voltage (V)	Resistance (kΩ)
20	0.97	12.0	12.4
40	2.22	12.0	5.41
60	4.80	12.0	2.50
80	9.23	12.0	1.30
100	17.14	12.0	0.70



Examiner
only

(a) (i) Plot the data on the grid below and draw a suitable line.

[4]



(ii) Describe the relationship between temperature and resistance.

[2]

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Examiner
only

- (b) (i) A resistor of resistance $5\text{ k}\Omega$ is now added to the original circuit, in parallel with the thermistor which is at a temperature of 30°C . The same 12 V power supply is used. Use your graph and equations from page 2 to calculate the total current in the circuit. [5]

Current = A

- (ii) Kim suggests that the current through both the resistor and the thermistor will decrease as the thermistor cools down. Explain whether or not she is correct. [3]

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